# Tech Manual
# TQ Series RO Systems



**Models:**

TQ 172
TQ 172N
TQ 173
TQ 173N
TQ 174
TQ 176

# TABLE OF CONTENTS



# TQ REVERSE OSMOSIS SERIES

**Manual part number: 10152B**


KINETICO®
Quality Water Systems

# About this Manual

This manual will cover information needed for the proper installation and operation of your Kinetico TQ Series Commercial Reverse Osmosis System. We have also included information regarding the frequently asked questions about reverse osmosis systems. This information may be more technical in nature, but provides further insight to the continued operation of this equipment to its highest standards.

This manual will use various icons to help highlight issues that are relevant to the safe operation of this equipment. The following icons will be used as described:

General information regarding the application of this product will be highlighted by this icon. This will include technical specifications and expected operational results.

**Lock out electrical power.**
Use appropriate lockout procedures when servicing.

**Maintain safe pressure.**
This sign indicates the safe operating pressure range.

**Consult Maintenance Section.**
Refer to the maintenance section for specific instructions.

**Consult Equipment Specifications Section.**
Refer to the equipment specifications section for specific instructions.

**Consult MSDS Sheets.**

A caution icon will be used to present any information that may hold a potential hazard or concern during the installation, use or maintenance of this product. **Should this information not be followed, it may result in damage of this equipment and its surroundings.**

**Electrical shock or electrocution hazard.**

**Pinch point or crushing hazard.**

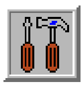
**Chemical hazard.**

The warning icon will be used to present any information that may result in a severe hazard or concern during the installation, use or maintenance of this product. **Should this information not be followed, it may result in severe physical harm.**

**Stay Clear.**

**Do Not Touch.**

**No Access.**
Only properly trained and authorized persons can enter area or open panel.



Any tools or materials required during the installation, use or maintenance of this equipment will be preceded by this icon. Using these specific tools will minimize time and effort. Not using the proper tool may result in damage to this equipment, its surroundings or even physical harm.

If there are any additional questions pertaining to this equipment, please contact your local Kinetico Dealer for further assistance.

## The Kinetico TQ RO Product Line

Kinetico's TQ product line provides both reverse osmosis and nanofiltration products that include electric and non-electric models. The reverse osmosis product line is designed to provide high quality reverse osmosis water using line pressure to operate all system features for the TQ 172 and TQ 173. The TQ 174 and TQ 176 use a pump to boost pressure to the system. Both the electric and non-electric versions are designed to allow the unit to be installed domestically and internationally, regardless of electrical requirements. The electric versions have a universal transformer that only requires a plug change to power the unit. The capacity of Kinetico's TQ RO Series systems accommodates daily production requirements from 150 to 450 gpd. The TQ Nanofiltration line is designed to provide a slightly lower quality permeate water than reverse osmosis. The capacity of Kinetico's TQ Nano Series accommodates daily production requirements from 200 to 300 gpd.

System features for the TQ Series include:

- Corrosion Resistant Aluminum Anodized Bracket
- Low Pressure / Low Temperature Production
- EverClean Rinse®
- Compact System Design
- Reverse Osmosis or Nanofiltration Membrane Choice TQ 172 and TQ 173

System Features for TQ 172 and TQ 173 (Wall Mount Systems)

- Non-Electric Operation / Optional booster pump available
- Wall-mountable, Space Saving Design
- Choice of RO or Nanofilter membranes

System Features for TQ 174 and TQ 176 (Cabinet Floor Systems)

- Integrated Low Voltage Pump
- Movable, Rolling Caster Design

A number of accessories are also available with the TQ Series systems including:

- Carbon Filters
- Backwashing Filters
- Softeners
- Storage Tanks

For further information regarding these products, please contact your local Kinetico Dealer.

**Wall Mount Systems**

| Model Name | TQ 172 | TQ 173 | TQ 172n | TQ 173n |
|---|---|---|---|---|
| Part Numbers | | | | |
| Reverse Osmosis System | 10162 | 10163 | XX | XX |
| Nanofiltration System | XX | XX | 10166 | 10167 |
| Rated* Daily Production | 150 gpd | 225 gpd | 200 gpd | 300 gpd |
| Rated* Daily Production | 568 lpd | 852 lpd | 757 lpd | 1,136 lpd |
| Membrane Type | Thin Film Composite | | | |
| Number of Membranes | 2 | 3 | 2 | 3 |

** Based on feed TDS of 500 mg/l and a water temperature of 77 °F*

**Floor Mount Systems**

| Model Name | TQ 174 | TQ 176 |
|---|---|---|
| Part Numbers | | |
| RO System 120v/60 hz (US Plug) | 11585 | 11560 |
| RO System 230v/50hz (European Plug) | 11589 | 11588 |
| Rated* Daily Production | 300 gpd | 450 gpd |
| Rated* Daily Production | 1,136 lpd | 1,704 lpd |
| Membrane Type | Thin Film Composite | |
| Number of Membranes | 4 | 6 |

** Based on feed TDS of 500 mg/l and a water temperature of 77 °F*

## System Sizing TQ 172 / 173

In the application of the TQ 172 / 173 Series wall mount systems, three important site variables must be taken into account:

Water Temperature
Inlet Pressure
Inlet Water Quality

To determine the output performance of the unit, a compensated inlet pressure must be determined. The inlet pressure is compensated based on the quality of the inlet water. For each 100 TDS, 1 psi is reduced for the actual inlet pressure.

**Example:**

| | |
|---|---|
| Actual Inlet Pressure | 55 psi |
| Inlet TDS | 1,000 mg/l |
| TDS Pressure Reduction: | 1,000 mg/l / (100 (mg/l)/psi) = 10 psi |
| Compensated Inlet Pressure: | 55 psi – 10 psi = 45 psi |

## TQ172

Reverse Osmosis System

Compensated Inlet Pressure

| Temperature | 40 psi | 50 psi | 60 psi | 70 psi | 80 psi |
|---|---|---|---|---|---|
| 40 °F | 33 gpd | 46 gpd | 55 gpd | 66 gpd | 71 gpd |
| 45 °F | 37 gpd | 52 gpd | 63 gpd | 74 gpd | 80 gpd |
| 50 °F | 42 gpd | 58 gpd | 71 gpd | 84 gpd | 90 gpd |
| 55 °F | 48 gpd | 66 gpd | 80 gpd | 95 gpd | 102 gpd |
| 60 °F | 54 gpd | 74 gpd | 90 gpd | 106 gpd | 114 gpd |
| 65 °F | 60 gpd | 83 gpd | 101 gpd | 119 gpd | 128 gpd |
| 70 °F | 67 gpd | 93 gpd | 113 gpd | 133 gpd | 144 gpd |
| 75 °F | 75 gpd | 104 gpd | 126 gpd | 149 gpd | 161 gpd |
| 80 °F | 82 gpd | 114 gpd | 138 gpd | 164 gpd | 176 gpd |
| 85 °F | 89 gpd | 123 gpd | 150 gpd | 177 gpd | 191 gpd |
| 90 °F | 97 gpd | 133 gpd | 162 gpd | 192 gpd | 207 gpd |
| 95 °F | 105 gpd | 144 gpd | 175 gpd | 208 gpd | 224 gpd |

## TQ172

Nanofiltration System

Compensated Inlet Pressure

| Temperature | 40 psi | 50 psi | 60 psi | 70 psi | 80 psi |
|---|---|---|---|---|---|
| 40 °F | 44 gpd | 61 gpd | 74 gpd | 87 gpd | 94 gpd |
| 45 °F | 50 gpd | 69 gpd | 84 gpd | 99 gpd | 107 gpd |
| 50 °F | 56 gpd | 78 gpd | 94 gpd | 112 gpd | 120 gpd |
| 55 °F | 64 gpd | 88 gpd | 106 gpd | 126 gpd | 136 gpd |
| 60 °F | 71 gpd | 98 gpd | 120 gpd | 142 gpd | 153 gpd |
| 65 °F | 80 gpd | 110 gpd | 134 gpd | 159 gpd | 171 gpd |
| 70 °F | 90 gpd | 124 gpd | 150 gpd | 178 gpd | 192 gpd |
| 75 °F | 100 gpd | 138 gpd | 168 gpd | 199 gpd | 214 gpd |
| 80 °F | 110 gpd | 152 gpd | 184 gpd | 218 gpd | 235 gpd |
| 85 °F | 119 gpd | 164 gpd | 200 gpd | 237 gpd | 255 gpd |
| 90 °F | 129 gpd | 178 gpd | 216 gpd | 256 gpd | 276 gpd |
| 95 °F | 140 gpd | 192 gpd | 234 gpd | 277 gpd | 298 gpd |

## TQ173

Reverse Osmosis System

Compensated Inlet Pressure

| Temperature | 40 psi | 50 psi | 60 psi | 70 psi | 80 psi |
|---|---|---|---|---|---|
| 40 °F | 50 gpd | 68 gpd | 83 gpd | 98 gpd | 106 gpd |
| 45 °F | 56 gpd | 77 gpd | 94 gpd | 111 gpd | 120 gpd |
| 50 °F | 63 gpd | 87 gpd | 106 gpd | 126 gpd | 135 gpd |
| 55 °F | 71 gpd | 99 gpd | 120 gpd | 142 gpd | 153 gpd |
| 60 °F | 80 gpd | 111 gpd | 134 gpd | 159 gpd | 172 gpd |
| 65 °F | 90 gpd | 124 gpd | 151 gpd | 179 gpd | 193 gpd |
| 70 °F | 101 gpd | 139 gpd | 169 gpd | 200 gpd | 216 gpd |
| 75 °F | 113 gpd | 155 gpd | 189 gpd | 224 gpd | 241 gpd |
| 80 °F | 124 gpd | 171 gpd | 207 gpd | 245 gpd | 264 gpd |
| 85 °F | 134 gpd | 185 gpd | 225 gpd | 266 gpd | 286 gpd |
| 90 °F | 145 gpd | 200 gpd | 243 gpd | 288 gpd | 310 gpd |
| 95 °F | 157 gpd | 216 gpd | 263 gpd | 312 gpd | 335 gpd |

## TQ173

Nanofiltration System

Compensated Inlet Pressure

| Temperature | 40 psi | 50 psi | 60 psi | 70 psi | 80 psi |
|---|---|---|---|---|---|
| 40 °F | 66 gpd | 91 gpd | 111 gpd | 131 gpd | 141 gpd |
| 45 °F | 75 gpd | 103 gpd | 125 gpd | 149 gpd | 160 gpd |
| 50 °F | 85 gpd | 117 gpd | 142 gpd | 168 gpd | 181 gpd |
| 55 °F | 95 gpd | 131 gpd | 159 gpd | 189 gpd | 204 gpd |
| 60 °F | 107 gpd | 148 gpd | 179 gpd | 213 gpd | 229 gpd |
| 65 °F | 120 gpd | 166 gpd | 201 gpd | 238 gpd | 257 gpd |
| 70 °F | 135 gpd | 185 gpd | 225 gpd | 267 gpd | 287 gpd |
| 75 °F | 150 gpd | 207 gpd | 252 gpd | 298 gpd | 321 gpd |
| 80 °F | 165 gpd | 227 gpd | 276 gpd | 327 gpd | 352 gpd |
| 85 °F | 179 gpd | 247 gpd | 299 gpd | 355 gpd | 382 gpd |
| 90 °F | 194 gpd | 267 gpd | 324 gpd | 384 gpd | 414 gpd |
| 95 °F | 209 gpd | 289 gpd | 350 gpd | 415 gpd | 447 gpd |

Note: The above charts are best on using an atmospheric storage or QuickFlo™ tank

# System Sizing TQ 174 / 176

In the application of the TQ Series cabinet systems, two important site variables must be taken into account:

**Water Temperature**
**Inlet Water Quality (TDS)**

**TQ 174 Production Chart 75 psi feed pressure to atmospheric tank**

| Temperature °F \ TDS | 500 | 1000 | 1500 | 2000 | 2500 |
|---|---|---|---|---|---|
| **45** | 189 | 169 | 151 | 133 | 116 |
| **55** | 240 | 214 | 189 | 166 | 142 |
| **65** | 300 | 267 | 234 | 203 | 174 |
| **75** | 371 | 328 | 287 | 247 | 209 |
| **85** | 445 | 391 | 340 | 291 | 245 |

Note: Production rates in gallons per day (GPD)

**TQ 176 Production Chart 70 psi feed pressure to atmospheric tank**

| Temperature °F \ TDS | 500 | 1000 | 1500 | 2000 | 2500 |
|---|---|---|---|---|---|
| **45** | 208 | 190 | 163 | 142 | 121 |
| **55** | 264 | 233 | 204 | 176 | 149 |
| **65** | 330 | 290 | 253 | 216 | 181 |
| **75** | 407 | 356 | 308 | 262 | 217 |
| **85** | 488 | 425 | 365 | 308 | 255 |

Note: Production rates in gallons per day (GPD)

**TQ 174 Production Chart 75 psi feed pressure to a bladder tank**

| Temperature °F \ TDS | 500 | 1000 | 1500 | 2000 | 2500 |
|---|---|---|---|---|---|
| **45** | 168 | 149 | 131 | 113 | 96 |
| **55** | 213 | 188 | 164 | 141 | 118 |
| **65** | 267 | 234 | 203 | 172 | 143 |
| **75** | 330 | 288 | 247 | 209 | 172 |
| **85** | 395 | 343 | 293 | 245 | 202 |

Note: Production rates in gallons per day (GPD)

**TQ 176 Production Chart 70 psi feed pressure to a Bladder tank**

| Temperature °F \ TDS | 500 | 1000 | 1500 | 2000 | 2500 |
|---|---|---|---|---|---|
| **45** | 183 | 160 | 139 | 118 | 98 |
| **55** | 232 | 202 | 173 | 146 | 120 |
| **65** | 290 | 252 | 214 | 179 | 145 |
| **75** | 358 | 309 | 261 | 216 | 174 |
| **85** | 430 | 368 | 309 | 254 | 203 |

Note: Production rates in gallons per day (GPD)

# EQUIPMENT SPECIFICATIONS

## System Configuration

Depending on the inlet characteristics of your water, additional components may be required with your TQ Series System. The configuration of these accessories along with their functionality has been listed at the end of this section. Should you require further information regarding this equipment, please contact your local Kinetico Dealer.

### Backwashing Filter

The TQ Series Systems require prefiltered water for adequate TSS (total suspended solids) removal. Kinetico offers a wide range of backwashing filters using a variety of medias. Depending on the types of solids in the water, the appropriate media is matched for your application. Kinetico's laboratory can test for the TSS content of feed water if it is not known to help determine the type of prefilter required. High levels of TSS can cause fouling of the TQ System membrane, thus reducing both the quality and quantity of the permeate water.

The maximum silt density index (SDI) for the influent water is 3.0. Operation above an SDI of 3.0 may damage membranes, pump or motor. At this point, the only remedy is to replace the damaged components.

### Carbon Filter

A carbon filter is required to protect the TQ System's membrane from organic fouling or from chlorine degradation. Both these types of fouling can seriously damage the TQ System's thin film composite membrane(s). With chlorine degradation, the membrane is slowly damaged. This type of damage is irreversible.

Each TQ unit is equipped with a sediment prefilter cartridge. For applications requiring carbon pretreatment where the actual daily consumption volume is low (< 50 gpd), a carbon sediment prefilter can be used in its place. One should note that the maximum capacity of this cartridge is 1,500 gallons at 2 mg/l of free $Cl_2$. Based on this capacity, cartridge replacement should be calculated.

The maximum influent level of chlorine to the TQ System is 0.05 mg/l. Prolonged exposure to excessive levels of chlorine will cause membranes to be destroyed. At this point, the only remedy is to replace the membranes.

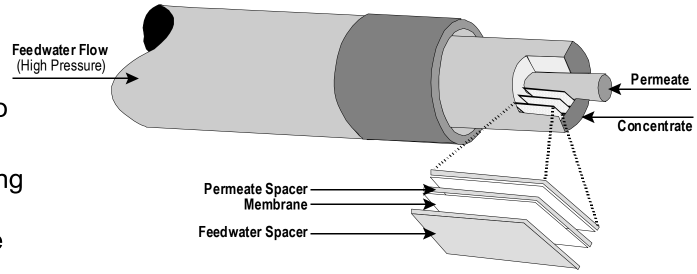
### Softener

For RO systems, a maximum inlet hardness of 4 gpg (68.4 mg/l as $CaCO_3$) is required. Above this level, scaling will occur within the RO's membrane(s). Calcium and magnesium are the two primary constituents leading to scale build up on the RO's membrane(s). To prevent this scaling, a softener can be used to remove calcium and magnesium from the water. A chemical descalent may be substituted for pre-softening the water in some applications. However, some chemical additives may result in a poorer quality of permeate water from the TQ RO.

For units with nanofiltration membranes, the maximum inlet hardness is 20 gpg. Nanofilter membranes are less sensitive to hardness fouling than their RO counterparts. A hard water source is preferred when specifying a nanofiltration unit, as the system's rejection rate for calcium and magnesium is much greater than it is for sodium ions.

## TQ Series Membranes

The TQ Series uses membrane technology to reduce the level of total dissolved solids (TDS) in a feed stream. The system uses spiral wound membranes for production of permeate water. The permeate water from the system typically exhibits a 95% or better reduction of TDS from the feed water when using reverse osmosis style membranes. With nanofiltration membranes, this rejection is typically on the order of 70% TDS reduction. The reject water contains the concentrated minerals that have not been permitted to pass through the membrane. The TQ system also features EverClean® Rinse which extends the operating life of the system's membranes by flushing the entire feed side of the membrane with high quality permeate water before its idling mode.




### Storage Tanks

Three different types of storage vessels are possible with the TQ Series.

### Pressure Storage Tank

These accumulator-type vessels come equipped with a valve, tee fitting and 1/2" tubing fitting. The tanks are precharged to 7 psi, allowing for the permeate water to remain pressurized during operation.

### QuickFlo® Storage Tank

Kinetico specially manufactures the QuickFlo® tank for operation with our membrane systems. The tank's special design allows for the unit to be vented to atmosphere while water is being processed to the tank. This configuration provides no backpressure, and therefore maximizes the output production of the membrane system. When water is required from the tank, an internal spool valve shifts, and the pressure from the tap source is now used to drive the water from the tank. During delivery, this configuration will provide a consistent flow and pressure.

### Atmospheric Storage Tank

These tanks offer the greatest storage volume. For applications that require a large volume of water, during a very small period of time, an atmospheric tank offers the best cost value.

## Pressurized Storage Tanks

| Model Number | Rated Size | Diameter | Height | Weight | Capacity to low level |
|---|---|---|---|---|---|
| 4461 | 10 gal. (39 l) | 16" (41 cm) | 24" 61 cm) | 24 lbs. (11 Kg) | 2.5 gal. (9.5 l) |
| 7481 | 20 gal. (78 l) | 16" (41 cm) | 33" (84 cm) | 26 lbs. (12 kg) | 4.8 gal. (18.2 l) |
| 7482 | 40 gal. (156 l) | 16" 41 cm) | 58" (147 cm) | 43 lbs. (20 Kg) | 9.7 gal. (36.7 l) |
| 7483 | 80 gal. (302 l) | 24" (61 cm) | 57" (145 cm) | 65 lbs. (30 Kg) | 16.4 gal. (62 l) |

* Empty Weight

## QuickFlo Tanks

| Model Number | Rated Size | Diameter | Height | Weight* | Capacity to low level |
|---|---|---|---|---|---|
| 9496 | 3 gal. (11.3 l) | 8" (20.3 cm) | 17" (43.2 cm) | 2.5 lbs. (1 Kg) | 3 gal. (11.3 l) |

## Atmospheric Tanks

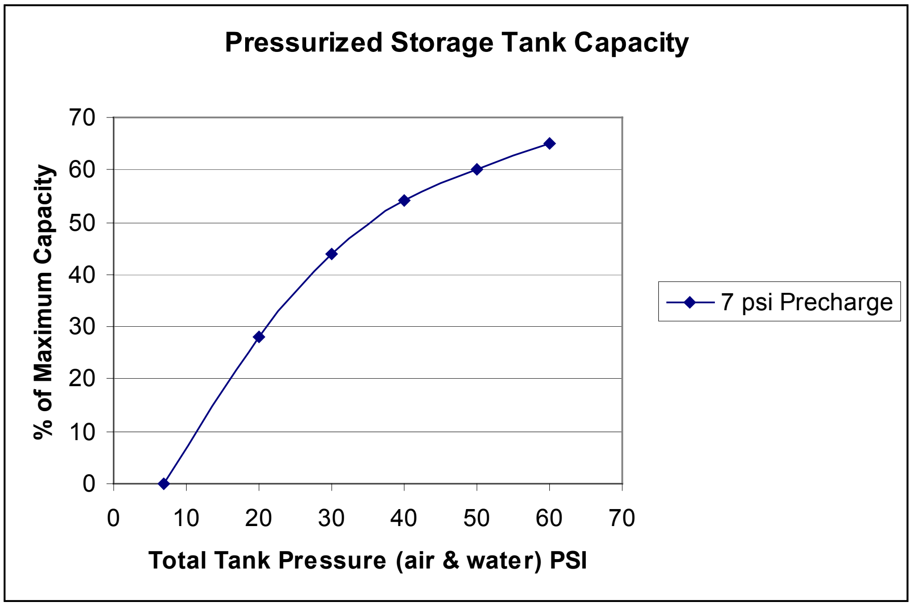
| Model Number | Rated Size | Diameter | Height | Weight* | Capacity to low level |
|---|---|---|---|---|---|
| 7495 | 300 gal. (1,136 l) | 35" (88.9 cm) | 80" (203.2 cm) | 85 lbs. (39 Kg) | 225 gal. (852 l) |
| 7496 | 550 gal. (2,082 l) | 48" (121.9 cm) | 83" 210.8 cm) | 120 lbs. (54 Kg) | 413 gal. (2,414 l) |
| 7497 | 850 gal. (3,218 l) | 48" (121.9 cm) | 120" (304.8 cm) | 190 lbs. (86 kg) | 638 gal. (2,840 l) |
| 7498 | 1,000 gal. (3,786 l) | 64" 162.6 cm) | 84" (213.4 cm) | 205 lbs. (93 Kg) | 750 gal. (2840 l) |

The chart below shows the storage capacity for pressurized storage tanks at various precharges.

To determine this volume:

1) Find the point on the horizontal axis at the bottom of the graph that represents the total pressure (air plus water) in the tank. Move vertically to the air pre-charge line found in step 1.

2) Move horizontally to the left axis and read the percent of “maximum total volume” in the tank under the above conditions. Multiply this percentage by the maximum total water volume to determine the volume of water in the tank.




## Repressurizer Pump

If an atmospheric-type holding tank is used, then the water must be repressurized for use throughout the facility.

## Atmospheric Storage Tank Accessories

| Part Number | Description |
|---|---|
| 7527 | Level Control Assembly Atmospheric Tank |
| 9837 | Bulkhead Kit, CRO |
| 10208 | Repressurizer, TS 115 Volt |
| 10209 | Repressurizer, TS 230 Volt |
| 8736* | 3 Gallon EverClean Rinse Accumulator |

* If an atmospheric tank is used with the level control, an accumulator tank must be added to shut off the unit properly.

The following fittings should be added:
#8939 Conn, Tube 3/8T x 1/2 Stem
#8308 Tee, 1/2 Union
#9336 Valve, Ball 3/8 Stem

## Filter Accessories

| Part Number | Description |
|---|---|
| 9046 | Filter Asy, Single 20" Housing w/Bracket |
| 9047 | Filter Asy, Double 20" Housing w/Bracket |
| 7474 | Cartridge, Carbon Matrix for 20" Housing |
| 367 | Cartridge, Prefilter 5 Micron for 20" Housing |
| 7477 | Cartridge, Sediment/Declor 5 Micron for 20" Housing |

## Miscellaneous

| Part Number | Description |
|---|---|
| 3814 | Wrench, Filter Sump |

# Operating Specifications

General operating parameters for the TQ product line have been listed in the table below. These specifications have been provided to express the required conditions for the operation of this system. If there are other parameters that you may be concerned about, please contact your local Kinetico Dealer for further operating or performance data.

Exceeding the conditions in the table below may inhibit performance or could cause the system to be permanently damaged.

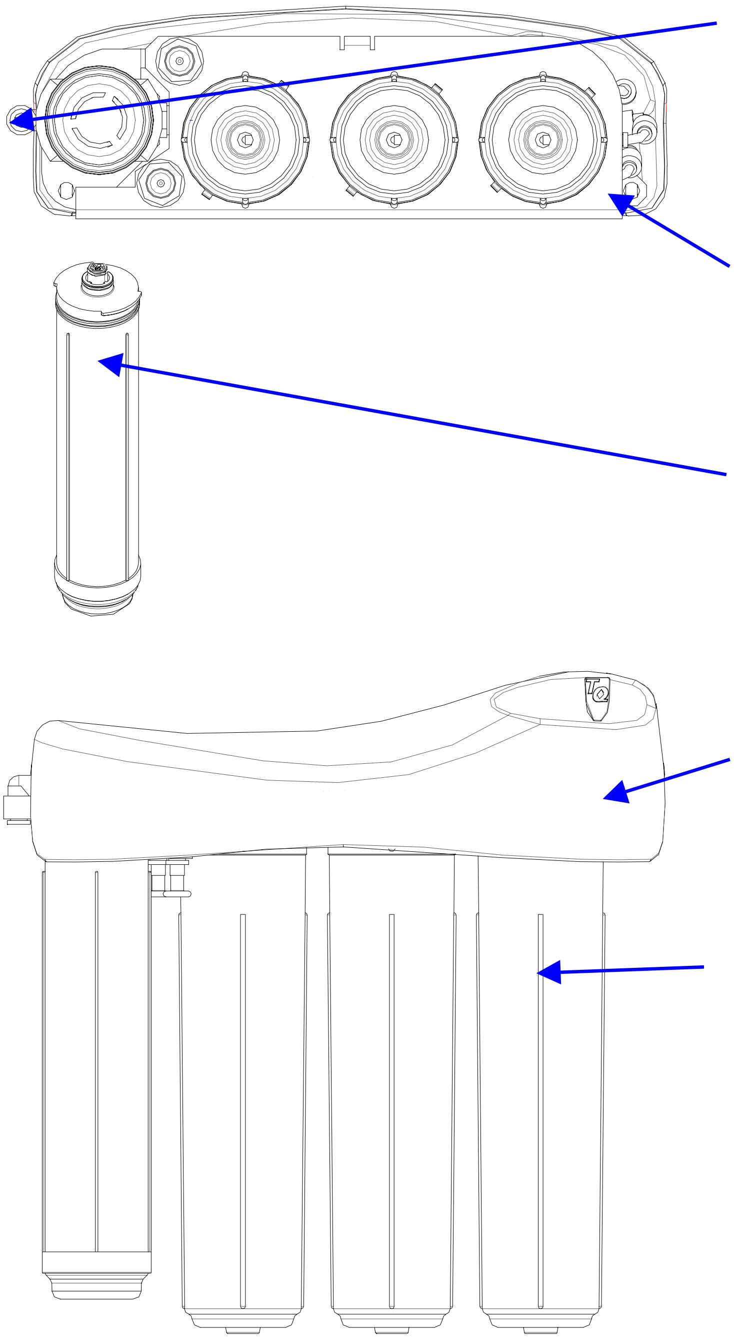
To assure that these factors are being met, a yearly water analysis is recommended to keep track of your influent water quality. This can be a beneficial tool in recognizing changes in your inlet water supply. If the inlet supply does change, please contact your area Kinetico Dealer. Modifications to the system or additional equipment may be required to maintain peak operation efficiency of your system.

| **Model #** | **TQ 172** | **TQ 173** | **TQ 172n** | **TQ 173n** | **TQ 174** | **TQ 176** |
|---|---|---|---|---|---|---|
| Inlet Filter | Sediment Cartridge | | | | None | |
| Membrane Performance | Low Energy RO or Nanoflitration | | | | Low Energy | |
| Membrane Type | Thin Film Composite | | | | | |
| Membrane Dimensions | 1.75" x 11" | | | | | |
| Number of Membranes | 2 | 3 | 2 | 3 | 4 | 6 |
| Square Footage of each Membrane | 1.6 $Ft^2$ | | | | | |
| Array Configuration | Parallel | | | | | |
| Drain Control | Capillary Tube | | | | | |
| Permeate Flush, per membrane | 5 oz reservoir | | | | | |
| System Shut-Off Control | Internal Pressure | | | | | |
| System Controller | Hydraulic | | | | | |
| Power | None | | | | 120/60 or 230/50 | |
| Inlet Connection | 3/8" Tubing | | | | | |
| Permeate Connection | 3/8" Tubing | | | | | |
| Drain Connection | 3/8" Tubing | | | | | |
| EverClean Rinse Connection | Internal/Automatic | | | | | |
| Pressure | 40 - 100 psi | | | | 20 - 80 psi | |
| Temperature | 40 - 95 $^oF$ | | | | | |
| pH | 4-11 SU | | | | | |
| Chlorine | < 0.05 mg/l as $Cl_2$ | | | | | |
| Hardness (Max.) | 4 gpg | 4 gpg | 20 gpg | 20 gpg | 4 gpg | 4 gpg |
| Silica (Max.) | < 10 mg/l as $SiO_2$ | | | | | |
| Iron (Max.) | < 0.1 mg/l as Fe | | | | | |
| Total Dissolved Solids (Max.) | < 2,500 mg/l as NaCl | | | | | |
| Inlet Flow Rate | 0.2 gpm | 0.3 gpm | 0.2 gpm | 0.42 gpm | .42 gpm | .62 gpm |
| Permeate Flow Rate | 0.1 gpm | 0.15 gpm | 0.1 gpm | 0.21 gpm | .21 gpm | .31 gpm |
| Reject Flow Rate | 0.1 gpm | 0.1 gpm | 0.1 gpm | 0.21 gpm | .21 gpm | .31 gpm |
| Daily Production* | 150 gpd | 225 gpd | 200 gpd | 300 gpd | 300 gpd | 450 gpd |
| Operating Pressure Range | 40 - 100 psi | | | | 20 - 80 psi | |
| Normal Operating Pressure | 70 psi | | | | 70 - 90 psi | |
| Rejection Rate - NaCl | > 95% | > 95% | > 50% | >50 % | > 95% | > 95% |
| Rejection Rate - $Ca_2SO_4$ | > 99% | > 99% | > 85% | >85 % | > 99% | > 99% |
| Recovery Ratio Atmospheric | 50% | 50% | 50% | 50% | 50% | 50% |
| EverClean Rinse Volume | 10 oz. | 15 oz. | 10 oz. | 15 oz. | 20 oz. | 30 oz. |
| Width | 18" | | | | 8.5" | |
| Depth | 6" | | | | 18.75 | |
| Height | 18" | | | | 20.5 | |
| Weight | 10 lbs. | 13 lbs. | 10 lbs. | 13 lbs. | 50 lbs. | 58 lbs. |
| **Part #** | 10162 | 10163 | 10166 | 10167 | 11585 | 11560 |

* Based on 77 degree F water, 500 TDS feed, Permeate to Atmospheric Tank, 70psi operating pressure

# Component Functionality TQ 172 - 173

The TQ System is comprised of various components. A brief description has been provided for each of these major components. The illustration helps depict the physical location of these components on the TS system, while the flow schematic shows the flow path through the system.



**Inlet** The inlet to the system is located on the left side of the TQ System. This connection is designed for 3/8" O.D. tubing.

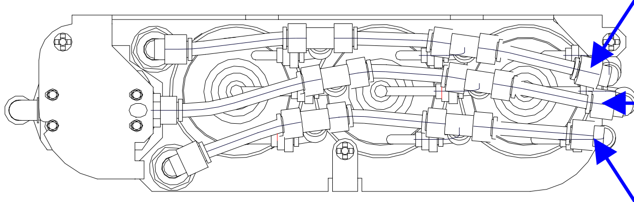
Do not use smaller than 3/8" tubing on the inlet connection.

**Bracket**
An aluminum anodized bracket is used to support the multi-membrane structure. The frame allows for quick and convenient installation in a wall-mounted configuration.

**Pre-filter Cartridge**
A sediment prefilter is included standard with each TQ system. If the daily water usage is low (<50 gpd), then a carbon prefilter could possibly be substituted for the sediment filter. Careful attention must be made to assure timely replacement of the carbon prefilter, due to its capacity for chlorine removal.

**System Cover**
A decorative cover is used to shield the feed, permeate and drain plumbing manifolds. This cover can be removed should maintenance be required on any of the internal components.

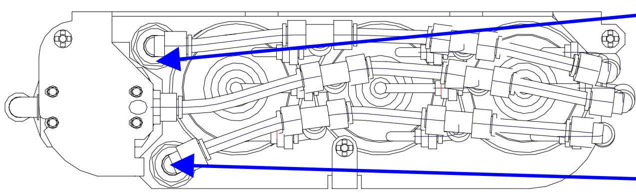
**Membrane Housing/Membrane**
Three membrane housings are used on the TQ Series. With the 172 configuration, only two of the housings include the membranes. This allows for a quick upgrade to a third membrane should the customer require the additional capacity. Membranes used are thin film composite. The operating efficiency of this membrane is affected by operating pressure, inlet water quality, water temperature and storage tank choice.



**Permeate Manifold** The permeate manifold (Blue) uses 3/8" tubing to connect water from the permeate to a common bulkhead.

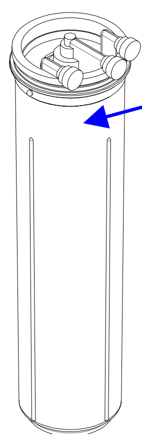
**Inlet Manifold** The inlet manifold (Black) uses 3/8" tubing to connect water from the outlet of the filter cartridge to each membrane housing.

**Drain Manifold** The drain manifold (Green) uses 3/8" tubing to connect water from the drain outlet of the membrane housings to the bulkhead.



**Permeate Outlet** The permeate outlet uses 3/8" tubing connection and is located at the back of the unit.

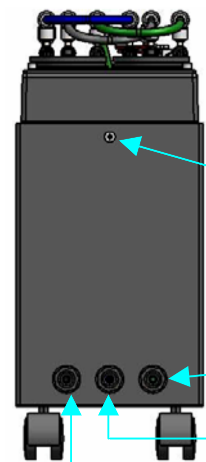
**Drain Outlet** The drain outlet uses 3/8" tubing connection and is located at the front of the unit.



**EverClean® Rinse** Each housing internally contains a permeate reservoir. During shut down modes, this water is used to flush the membranes, keeping them in optimal condition.

# Component Functionality TQ 174 - 176

The TQ System is comprised of various components. A brief description has been provided for each of these major components. The illustration helps depict the physical location of these components on the TS system, while the flow schematic shows the flow path through the system.



Connections: All connections are located on the back of the system.

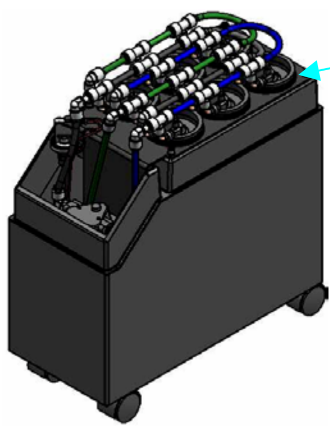
Do not use smaller than 3/8" tubing on any connection.

Power cord plug – 24 vdc

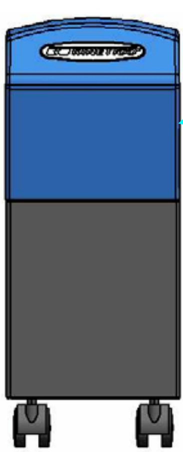
Drain Bulkhead 3/8" Push fit

Permeate Bulkhead 3/8" Push fit

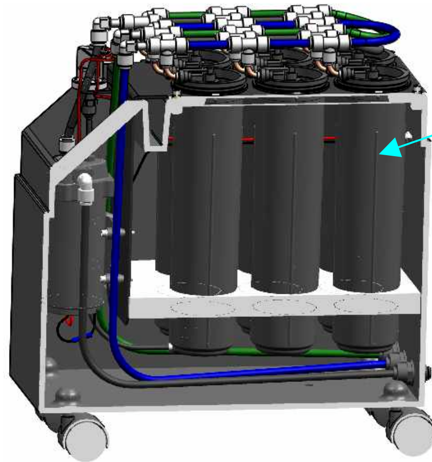
Inlet Bulkhead 3/8" Push fit



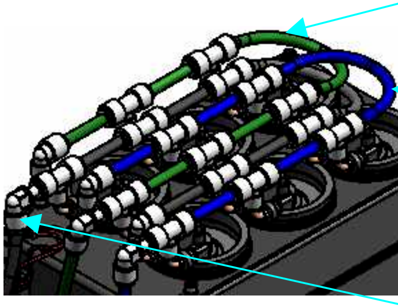
**Bracket** An aluminum anodized bracket is used to support the multi-membrane structure and the system pump.



**System Cover** A decorative cover is used to shield the feed, permeate and drain plumbing manifolds. This cover can be removed should maintenance be required on any of the internal components.



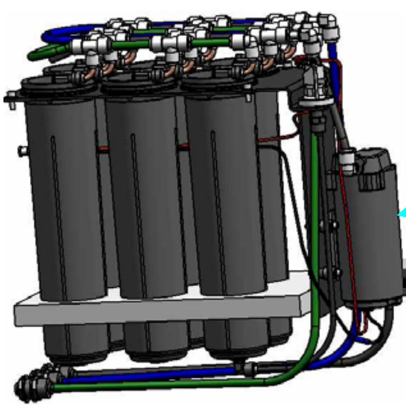
**Membrane Housing/Membrane** The cabinet houses four or six membrane housings. Four housings are used on the 174 and six on the 176. This allows a 174 to be upgraded to a 176 should the customer require the additional capacity. Membranes used are thin film composite. The operating efficiency of this membrane is affected by operating pressure, inlet water quality, water temperature and storage tank choice.



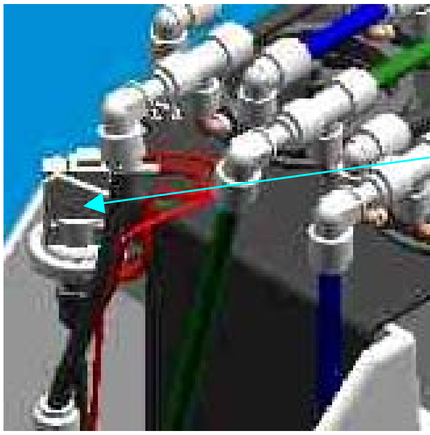
**Drain Manifold** The drain manifold (Green) uses 3/8" tubing to connect water from the drain outlet of the membrane housings to the bulkhead.

**Permeate Manifold** The permeate manifold (Blue) uses 3/8" tubing to connect water from the permeate to a common bulkhead.

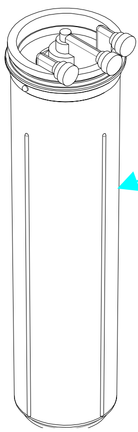
**Inlet Manifold** The inlet manifold (Black) uses 3/8" tubing to connect water from the outlet of the filter cartridge to each membrane housing.



**System Booster Pump** The pump is located inside the cabinet on the bracket. It is used to boost the system pressure on the feed to the membranes.



**Pressure Switch** The inlet pressure switch is used to start the booster pump. It is preset and does not need to be adjusted.



**EverClean® Rinse** Each housing internally contains a permeate reservoir. During shut down modes, this water is used to flush the membranes, keeping them in optimal condition.

# INSTALLATION

## Getting Started

The following procedures have been developed to assist during the installation of your TQ System.

The installation of this TQ Membrane System should be performed by a qualified service person with an understanding of local and regional codes that may affect the installation requirements.

## Pre-installation Review

Should there be any visible shipping damage to the TQ System, please contact your freight company for coverage information. Before beginning the installation of the TQ System, confirm system configuration to be installed, and components that have been ordered. Please review TQ System specification sheet that includes required components.

Review of the customer's facility is also recommended, especially critical operating data that could effect the operation of the system:

**Water Pressure**
Water pressure to the TQ System will effect maximum flow permitted by the system. The TQ 172 – 173 systems will not operate if the inlet pressure fluctuates below a dynamic pressure of 40 psi. This minimum pressure must be maintained to the system at all times. Should the pressure fluctuate below this level, a booster pump may be required. The TQ 174 – 176 require 20 psi. inlet pressure.

**Temperature**
Ambient temperature must be maintained above 32°F (0°C). Freezing temperatures will cause breakage of equipment and void all warranties.

**Water Temperature**
Inlet water temperature must be maintained between 40°F (4.4 °C) and 95°F (35 °C) to prevent damage to the system's membranes.

**System Location**
The unit must be installed indoors. Failure to comply with this requirement can cause significant damage to the system and will create a safety concern.

## TS RO Installation

**Tools and Installation Materials**
Since the TQ System process high quality water, plumbing runs on the process, purge and drain outlets should all be completed with High Density P.P or P.E. tubing. Copper and galvanized pipe will be chemically attacked by the permeate water.

Teflon™ Tape
Fitting Sealant
3/8" Tubing Wall Mount

Plastic Tube Cutter
Phillips head screwdriver (medium and small)
Additional 316 SS pressure gauges

## Unpacking

Remove the TQ System and components from the shipping carton and check for damage.
When handling the wall unit, pick it up by the metal frame. Lift the cabinet unit by lifting the cabinet. Lifting the system by any other component may damage the unit.

Check all tube fitting connections and make sure they are tight.

## Cover

Install the cover to the top of the unit.

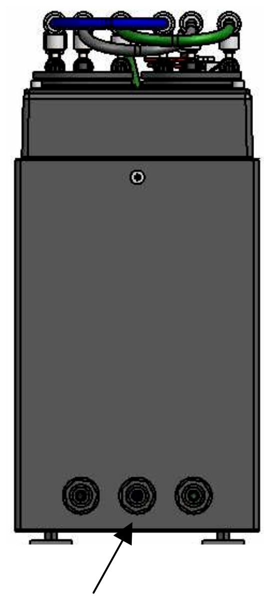
## System Hook-ups

1. The wall mount system should securely mounted to the wall. Two 1/4” diameter mounting holes, 8” on center, are provided on the back of the system. The cabinet unit sits on the floor. Rolling casters can be added to the bottom as an option.

2. Connect the water supply to the feed port on the left side of the wall mount unit (when facing the unit). One 3/8” O.D. tubing will be needed for this connection. The cabinet unit feed port is located on the back side of the cabinet. One 3/8” O.D. tubing will be needed for this connection. Be sure to install a shutoff valve on the inlet supply.

3. To avoid excessive pressure drop to the system, be sure the inlet tubing run is kept to a minimum. (Recommended less than 10’.) If a longer plumbing run is required, use a large size plumbing, then reduce the size when the line gets to the TQ System.

4. Connect the permeate outlet to the storage tank. This connection is made with 3/8” tubing. A relief valve may be required between this connection.

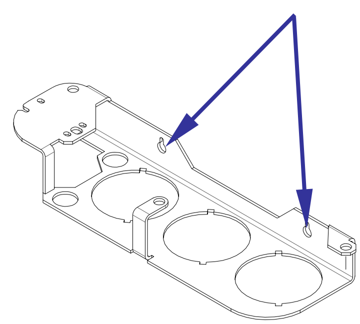



5. Connect the drain port to a suitable drain outlet. Be sure to follow applicable plumbing codes when making the drain connection. Use 3/8” tubing for the connection.

6. Open the feed water supply valve slowly. Check the system for leaks.

7. Disconnect the permeate line from the storage vessel and temporarily connect to a drain or five gallon bucket so the preservative can be flushed out of the system.

8. Plug the cabinet unit into a standard 115 / 60hz or 230 / 50hz electrical outlet. Use the power cord supplied with the unit.



# OPERATION AND MAINTENANCE

## Normal Operating Procedures

During the normal operation of the TQ System, a few items are required to be checked daily. Routine upkeep with these items will maintain the operation of the TQ to its highest standards.

### Daily Start-up/Shutdown

The equipment will run in an automatic mode, shutting itself off when the storage tank is full. Each time the unit shuts down, the EverClean® Rinse is set to flush each membrane with permeate water.

### Daily Log

On the back page of this manual, a daily log has been provided. Photocopy this template and use it for recording the system's daily performance. By recording various system characteristics each day, this will help in any future troubleshooting of the equipment. It is also necessary to keep a daily performance log to show changing performance characteristics of the TQ System. These changes may be the result of membrane fouling. With accurate logs kept on performance, timely cleaning of the membranes will insure maximum membrane life.

The parameters included on the daily log sheet are:

Date
Feed Pressure (if an extra gauge is added)
Permeate Flow Rate
Water Temperature
Water Quality
Comments

### Routine Maintenance

**Filter Maintenance (Optional)**

The inlet filter should be change at least twice per year. If the filter becomes blocked, the dynamic pressure to the membranes will be reduced, and the system's performance will be diminished. The system will not operate properly if the inlet pressure drops below 40 psi for the wall-mount unit (20 psi for cabinet unit). The cabinet unit does not have a prefilter standard, so one must be added.

Check the daily log for changes in the inlet pressure if an optional gauge is added. One can predict the life of the filter before its use has become over-exhausted by using the daily log as a preventative maintenance tool.

If a carbon cartridge is used with the system, then cartridge replacement will become more frequent. The capacity of the carbon cartridge is 1500 gallons, based on 2 mg/l of free $Cl_2$. Cartridge replacement must occur before exhaustion, or the membrane will be damaged.

## System Commissioning (Initial Start-up)

The start-up procedures for the equipment should be followed if:

-It is the first time the unit is being put into operation.
-The equipment has been moved.

-The unit has been shut down for an extended period of time (a few weeks).

After completing these start-up procedures, the normal operating procedure should be followed.

1. Make sure the unit has been properly installed by reviewing the installation section.

2. Temporarily connect the permeate line to a drain or a bucket.

**IMPORTANT**
**The membrane sanitizing solution must be properly rinsed from the membrane during installation. DO NOT drink the water from the system during this rinsing procedure.**

3. Open the feed water valve slowly to pressurize the system. (For TQ 174 & TQ 176 plug unit into electrical connection.)

4. Check the unit for leaks.

5. Tighten any connections exhibiting leaks.

6. Do not exceed 100 psi on system. Excessive pressure will damage the system.

7. After 1/2 hour of operation, the membranes should be adequately flushed of any preservatives.

8. Shut the system down by closing the inlet water valve.

9. Connect the permeate line to the storage tank.

10. Restart the system by opening the inlet water valve.

11. At this time, permeate water should begin filling the storage tank.

## Sanitization Procedure for Standard storage tank

a. Shut off the inlet water supply to the TQ system.

b. Drain any remaining water from the storage tank.

c. Connect a sanitization cartridge in the line feeding the storage tank. The sanitization cartridge is a separate device that must be plumbed in and removed after the sanitization is complete. Connect a line from the inlet water supply to the inlet of the sanitation cartridge. Connect a line from the outlet of the sanitation cartridge to the storage tank.

d. Fill the sanitization cartridge 3/4 full with clean water. Add (2 ml) of regular, unscented household bleach (5-1/4% sodium hypochlorite) to the center of the cartridge for every gallon the storage tank holds (example a 20 gallon tank would have 40ml added). (If a sanitization cartridge is not avaliable, household bleach may be added directly into the tubing feeding the storage tank.)

e. Be sure the storage tank valve is open and the outlet valve is closed. Slowly open the cold water supply valve and allow chlorinated water to fill the storage tank (fill time may vary based on the size of the storage tank).

f. Close the cold water supply valve and the storage tank valve. Open the outlet valve to depressurize the system, then close the valve. Remove the sanitization cartridge and discard the water it contains. *Be* *prepared to catch accidental spills from this cartridge.* Install the protective cap back on the sanitization cartridge, and save it for future use.

g. Reconnect the tubing to the storage tank.

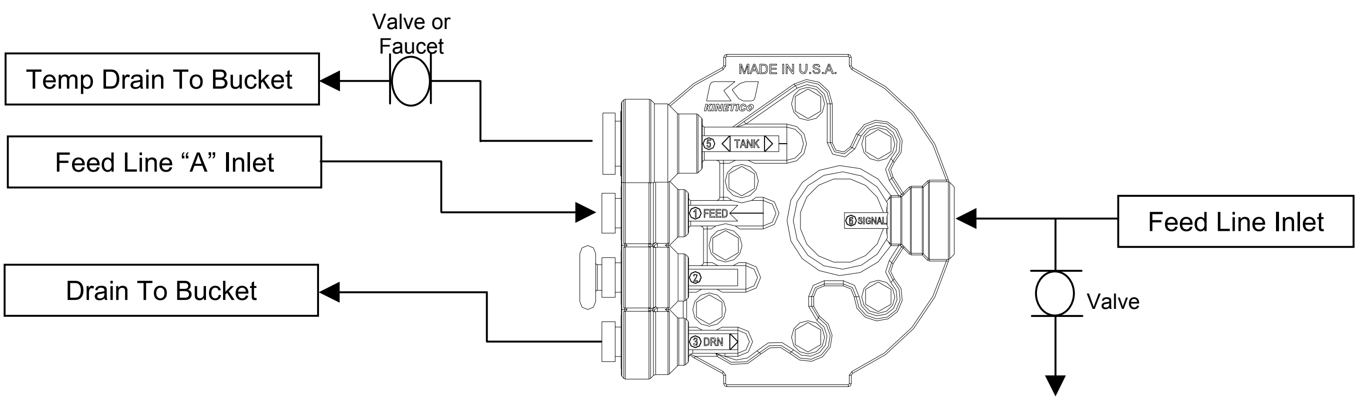
h. Open cold water supply valve and open ball valve on the storage tank. This will allow the system to fill the storage tank with water. After the storage tank has filled, discard the water in the tank. When the tank refills for the second time the system is ready for use.

## Sanitization Procedure when using QuickFlo Storage Tank

The following steps outline the sanitizing and/or pre-filling procedure of a QuickFlo® tank prior to installation. *NOTE: It is recommended that this procedure be performed off site.*

**Requirements**:

- Two pressurized feed lines.
- One line that can introduce chlorinated water to the product side of the tank.




## Membrane Sanitization

a. Shut off the inlet water supply to the TQ system.

b. Drain any remaining water from the storage tank.

c. Remove the prefilter cartridge. Be prepared to catch accidental water spills when removing the cartridge.

d. Use a sanitization cartridge to sanitize the membrane. Add 1 cup of clean water with 3 teaspoons of Iron Out or sodium meta bisulfite to the center of the sanitization cartridge. It may be best to premix the solution first to get an adaquate dilution before adding it to the sanitization cartridge. A small, plastic, household funnel or the dropper provided in Kinetico's sanitization kit may aid in adding solution to the cartridge. After adding solution to the cartridge, install cartridge to the prefilter head.

e. Disconnect the tubing from the storage tank ball valve, and put the tubing end into a bucket or drain. Open the cold water supply valve, and allow the product water to discharge from the tubing into the bucket or drain for 15 to 20 minutes. Close the cold water supply valve, and reconnect the tubing to the storage tank ball valve. If the tubing end is scarred or scratched, cut away the damaged end before connecting it to the tank.

f. Remove the sanitization prefilter cartridge, and replace it with a new prefilter cartridge.

g. Open ball valve on the tank and the cold water feed line. Allow the system to fill the storage tank for several hours. Discard the water in the tank after this time. When tank refills, the system is ready to use.

# Troubleshooting

**Poor Quality**

| | |
|---|---|
| **Inlet TDS too high** | **Check inlet TDS. TQ RO is designed to operate with a ~ 98% rejection. The TQ Nano is designed to operate with a ~ 85% rejection.** |
| **Permeate flush not operating** | **Cycle system through permeate flush by allowing the storage tank to fill and go through a normal flush cycle. Check flush time and volume to reject during flush. Check reject quality at the end of the flush; it should be RO quality.** |
| **System not operating within specifications** | **Confirm system operating parameters see Operating Specifications table. The system should run at provided permeate and reject flow rates. Also check the system pressure.** |
| **Brine seal leak** | **Check permeate conductivity for each housing. If one or some show poor readings, service the membranes.** |
| **Chlorine/Oxidizer damage to membranes** | **Check inlet chlorine or ozone levels. Residual less than 0.05 mg/l is required to prevent thin film composite membrane deterioration. Check downstream of the prefilter.** |
| **Pressure too low (<40psi)** | **Check and add booster pump if pressure is too low.** |
| **Membrane plugged or fouled** | **Check, clean or replace.** |
| **Insufficient reject flow rate** | **See "No reject water."** |
| **Membrane expended** | **Replace membrane.** |

**System Not Consistently Running**

| | |
|---|---|
| **Low permeate flow** | **Water temperature is low. Plugged prefilter or low feed psi.**<br><br>**Membranes are fouled.** |
| **Membranes fouled** | **Check system log for history. If consistent declining production is found, clean membranes. Consult your Kinetico dealer for appropriate cleaning procedures and chemicals.**<br><br>**Confirm proper system operating parameters.** |

| | |
|---|---|
| **No accumulator tank** | **A stable pressure signal is required for the unit to shut off properly. In some cases, only using tubing to connect the TQ unit to an atmospheric storage tank may not provide enough volume in the line to hold the pressure signal. The addition of an accumulator tank (less than 1 gallon) will resolve this issue.** |
| **Accumulator tank lost air precharge** | **Pump up tank air pressure to 7 psi. Check for cause of failure.** |
| **Prefilter fouled** | **Replace.** |

**Frequent Membrane Cleaning Required**

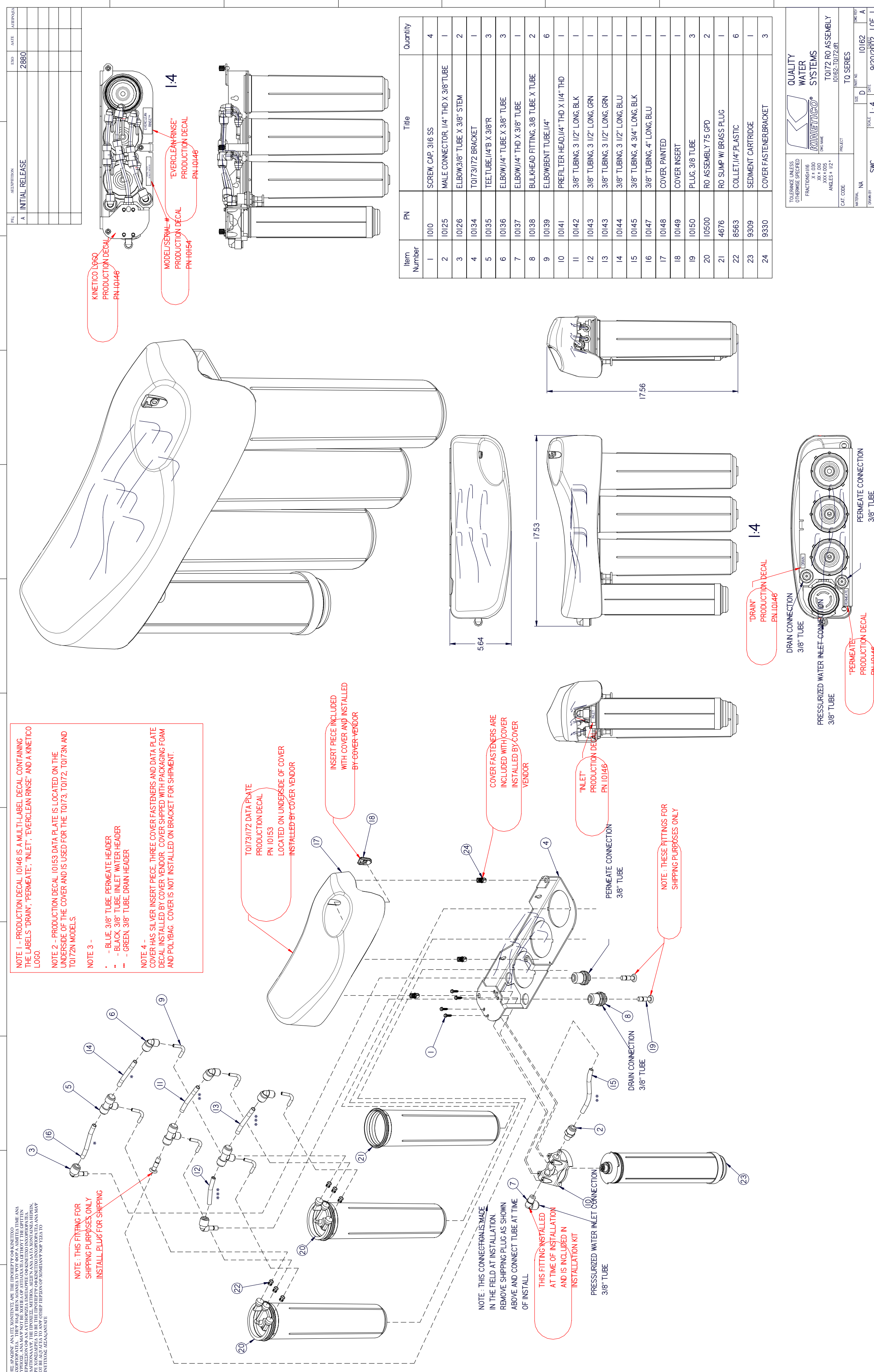
| | |
|---|---|
| **Ineffective cleaning solution** | **Silica treatment requires caustic cleaners that are heated to 90$^{o}$F for maximum effectiveness. Hardness contamination is cleaned with acid chemistries.** |
| **Improper cleaning procedures** | **Temperature, flow rate, time and chemistry must all be matched to the specific foulant being cleaned. Improper procedure will result in poor cleaning cycles.** |
| **Damaged membranes** | **Replace membranes.** |

**Poor Production Volume**

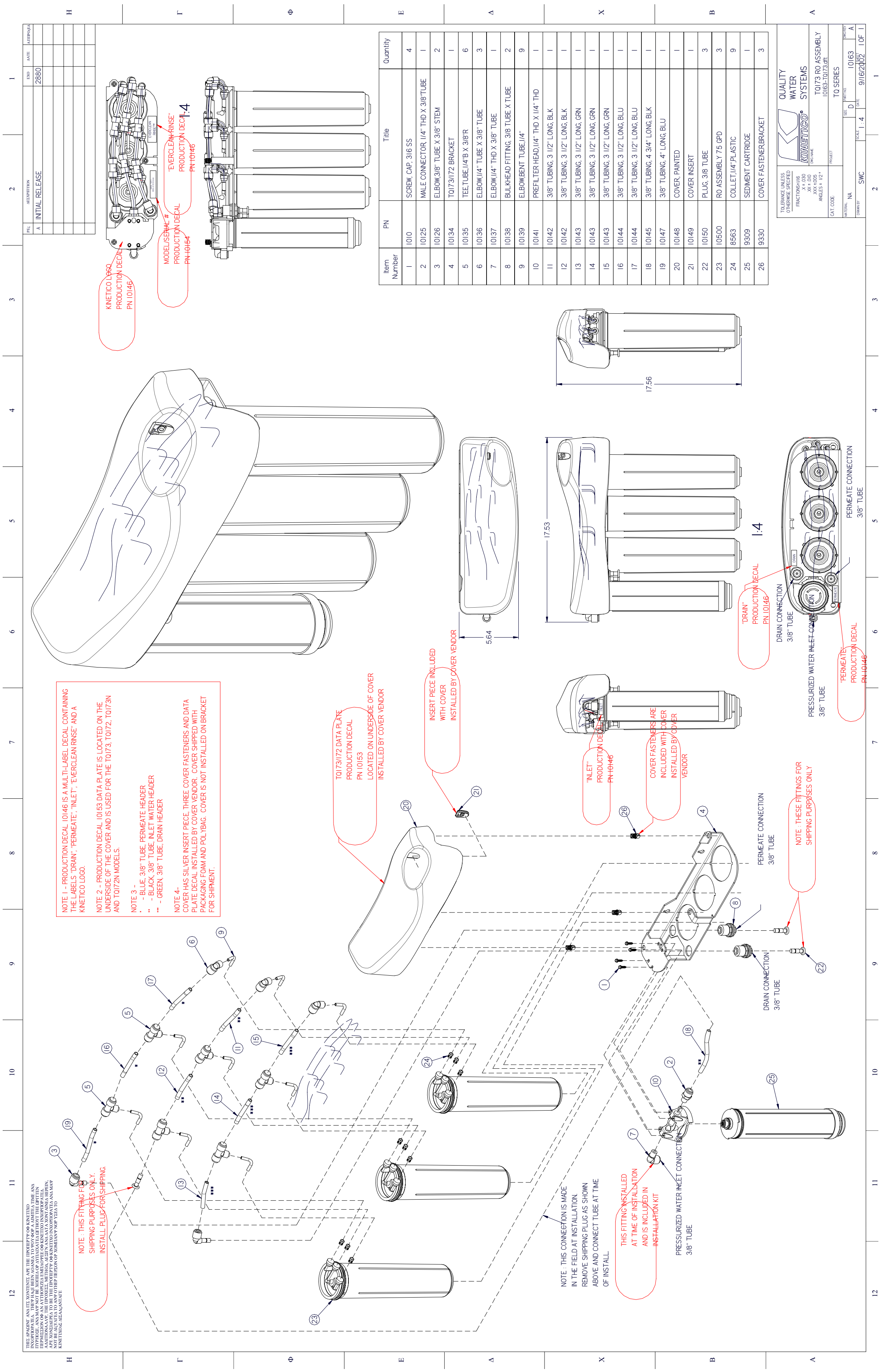
| | |
|---|---|
| **System not operating within specifications** | **Confirm system operating parameters. See Operating Specifications table. The system should run at provided permeate, and reject flow rates. Also check the system pressure.** |
| **Cold Temperature** | **Inlet temperature is less than 77$^{o}$F. Cold water will have a lower production rate. See sizing charts for different temperatures.** |
| **Fouled membranes** | **Check permeate quality with a hand-held meter. Also check permeate flow rate.** |
| **Inlet quality changes** | **Analyze the TQ Systems' feed for chlorine, hardness, temperature and SDI.** |
| **Tank precharge pressure too high** | **Relieve to 7 psi. Check pressure while system is empty.** |
| **Tank lost air precharge** | **Pump up air precharge to 7 psi and check for cause of failure. Note: tank must be empty to add more precharge.** |
| **Prefilter cartridge plugged** | **Replace cartridge.** |
| **Insufficient reject flow rate** | **See "No reject water."** |
| **Membrane fouled** | **Determine and correct cause; replace membrane. A new feed analysis may be required.** |
| **Insufficient storage capacity** | **Check charts in Options Section for draw down capacity.** |

| | |
|---|---|
| **No reject water** | |
| **Plugged drain line or outlet** | **Check and clean.** |
| **Low flow rate at point of use** | |
| **Low water production** | **See “No water or not enough water.”** |
| **Tank lost air precharge** | **Pump up air precharge to 7 psi. and check for cause of failure.** |
| **Tank diaphragm failed** | **Replace as necessary.** |
| **Pump pressure low** | **Check and adjust.** |
| **Bad tasting water** | |
| **Increase in product TDS** | **See “High product water TDS.”** |
| **Postfilter cartridges exhausted** | **Replace filter cartridges.** |
| **Tank contaminated** | **Re-sanitize tank/system, and replace post filter cartridge.** |
| **Unit cycles on/off intermittently** | |
| **Air precharge too low** | **Adjust to 7 psi while the tank is empty.** |

# Parts




| Item Number | PN | Title | Quantity |
|---|---|---|---|
| 1 | 1010 | SCREW, CAP, 316 SS | 4 |
| 2 | 10125 | MALE CONNECTOR, 1/4" THD X 3/8"TUBE | 1 |
| 3 | 10126 | ELBOW3/8" TUBE X 3/8" STEM | 2 |
| 4 | 10134 | TQ173/172 BRACKET | 1 |
| 5 | 10135 | TEE TUBE,1/4"B X 3/8"R | 3 |
| 6 | 10136 | ELBOW,1/4" TUBE X 3/8" TUBE | 3 |
| 7 | 10137 | ELBOW,1/4" THD X 3/8" TUBE | 1 |
| 8 | 10138 | BULKHEAD FITTING, 3/8 TUBE X TUBE | 2 |
| 9 | 10139 | ELBOWBENT TUBE,1/4" | 6 |
| 10 | 10141 | PREFILTER HEAD,1/4" THD X 1/4" THD | 1 |
| 11 | 10142 | 3/8" TUBING, 3 1/2" LONG, BLK | 1 |
| 12 | 10143 | 3/8" TUBING, 3 1/2" LONG, GRN | 1 |
| 13 | 10143 | 3/8" TUBING, 3 1/2" LONG, GRN | 1 |
| 14 | 10144 | 3/8" TUBING, 3 1/2" LONG, BLU | 1 |
| 15 | 10145 | 3/8" TUBING, 4 3/4" LONG, BLK | 1 |
| 16 | 10147 | 3/8" TUBING, 4" LONG, BLU | 1 |
| 17 | 10148 | COVER, PAINTED | 1 |
| 18 | 10149 | COVER INSERT | 1 |
| 19 | 10150 | PLUG, 3/8 TUBE | 3 |
| 20 | 10500 | RO ASSEMBLY 75 GPD | 2 |
| 21 | 4676 | RO SUMP W/ BRASS PLUG | 1 |
| 22 | 8563 | COLLET,1/4" PLASTIC | 6 |
| 23 | 9309 | SEDIMENT CARTRIDGE | 1 |
| 24 | 9330 | COVER FASTENERBRACKET | 3 |

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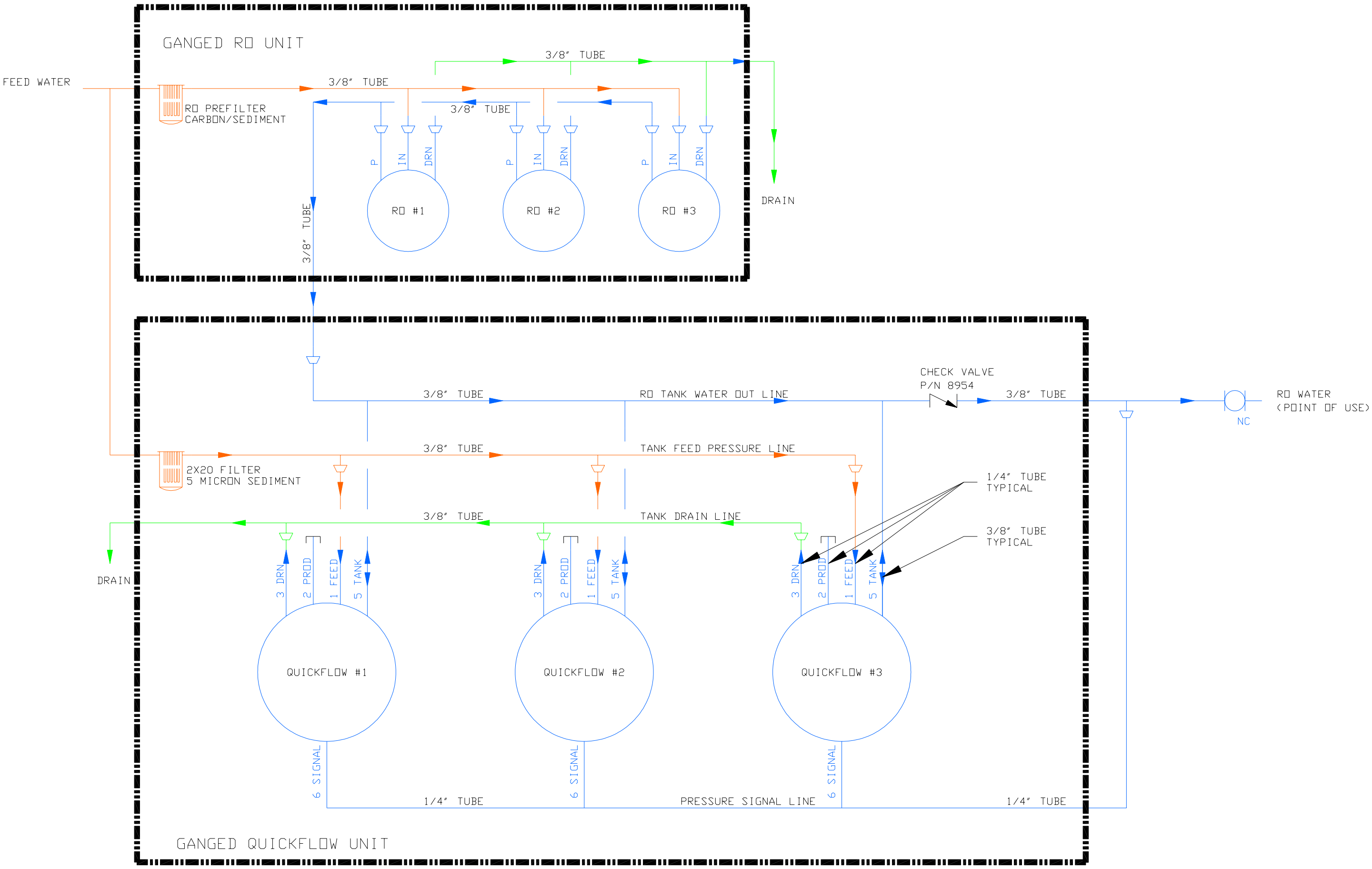
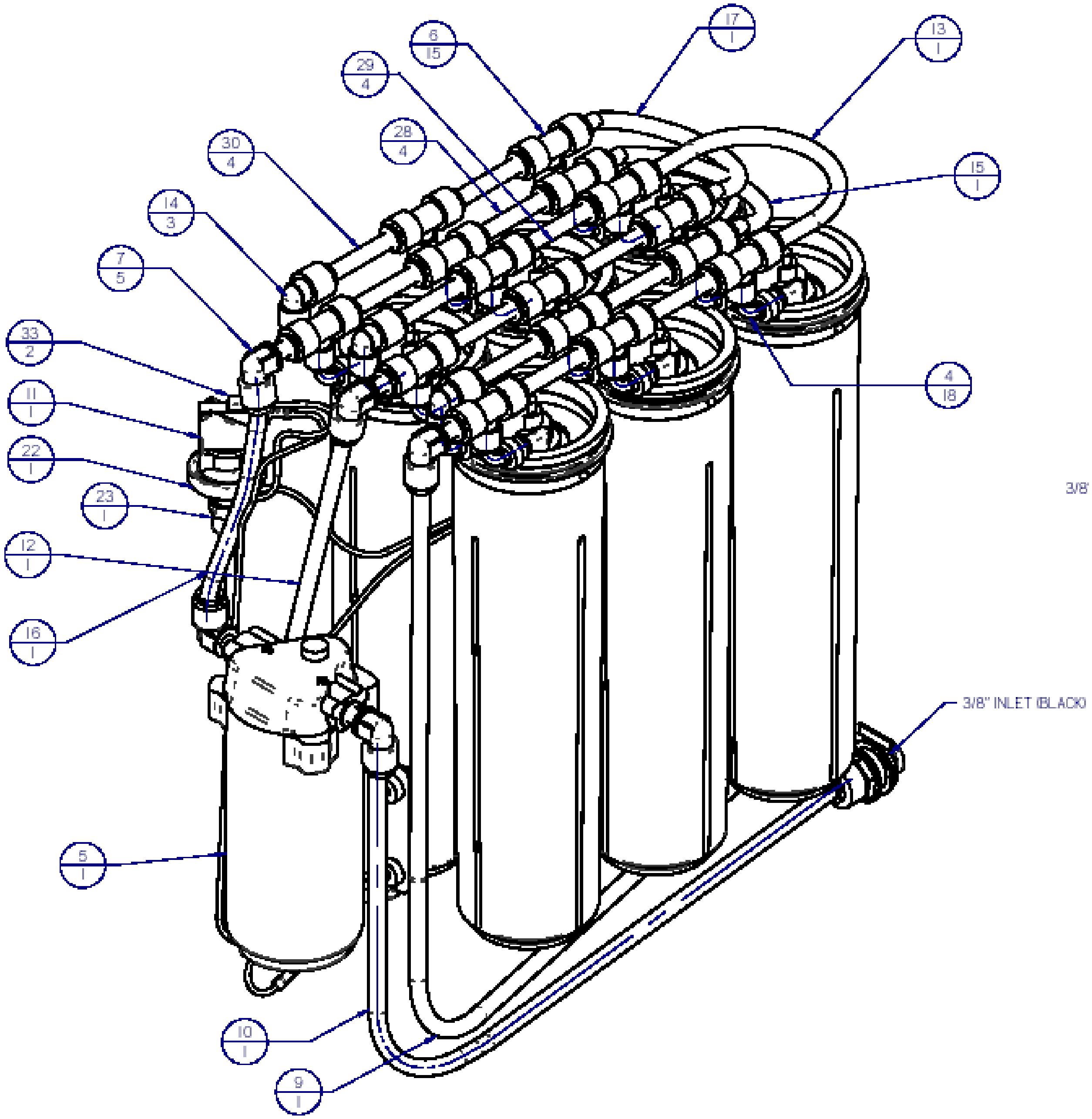
| 4 | 10134 | TQ173/172 BRACKET | 1 |
| 5 | 10135 | TEE,TUBE,1/4"B X 3/8"R | 6 |
| 6 | 10136 | ELBOW,1/4" TUBE X 3/8" TUBE | 3 |
| 7 | 10137 | ELBOW,1/4" THD X 3/8" TUBE | 1 |
| 8 | 10138 | BULKHEAD FITTING, 3/8 TUBE X TUBE | 2 |
| 9 | 10139 | ELBOW,BENT TUBE,1/4" | 9 |
| 10 | 10141 | PREFILTER HEAD,1/4" THD X 1/4" THD | 1 |
| 11 | 10142 | 3/8" TUBING, 3 1/2" LONG, BLK | 1 |
| 12 | 10142 | 3/8" TUBING, 3 1/2" LONG, BLK | 1 |
| 13 | 10143 | 3/8" TUBING, 3 1/2" LONG, GRN | 1 |
| 14 | 10143 | 3/8" TUBING, 3 1/2" LONG, GRN | 1 |
| 15 | 10143 | 3/8" TUBING, 3 1/2" LONG, GRN | 1 |
| 16 | 10144 | 3/8" TUBING, 3 1/2" LONG, BLU | 1 |
| 17 | 10144 | 3/8" TUBING, 3 1/2" LONG, BLU | 1 |
| 18 | 10145 | 3/8" TUBING, 4 3/4" LONG, BLK | 1 |
| 19 | 10147 | 3/8" TUBING, 4" LONG, BLU | 1 |
| 20 | 10148 | COVER, PAINTED | 1 |
| 21 | 10149 | COVER INSERT | 1 |
| 22 | 10150 | PLUG, 3/8 TUBE | 3 |
| 23 | 10500 | RO ASSEMBLY 75 GPD | 3 |
| 24 | 8563 | COLLET,1/4",PLASTIC | 9 |
| 25 | 9309 | SEDIMENT CARTRIDGE | 1 |
| 26 | 9330 | COVER FASTENER,BRACKET | 3 |


GANGED RO UNIT
FEED WATER
3/8" TUBE
RO PREFILTER
CARBON/SEDIMENT
3/8" TUBE
3/8" TUBE
P
IN
DRN
RO #1
RO #2
RO #3
DRAIN
3/8" TUBE
CHECK VALVE
P/N 8954
3/8" TUBE
RO TANK WATER OUT LINE
3/8" TUBE
RO WATER
(POINT OF USE)
NC
2X20 FILTER
5 MICRON SEDIMENT
TANK FEED PRESSURE LINE
1/4" TUBE
TYPICAL
3/8" TUBE
TYPICAL
TANK DRAIN LINE
DRAIN
3 DRN
2 PROD
1 FEED
5 TANK
QUICKFLOW #1
QUICKFLOW #2
QUICKFLOW #3
6 SIGNAL
1/4" TUBE
PRESSURE SIGNAL LINE
1/4" TUBE
GANGED QUICKFLOW UNIT

# Parts TQ 174/176

| Item # | Part # | Title | Qty |
|---|---|---|---|
| 1 | 11561 | Cabinet, TQ176 | 1 |
| 2 | 11574 | Bracket, TQ176 | 1 |
| 3 | 8563 | Collet, 1/4", Plastic | 18 |
| 4 | 10139 | Module Tube 90 elbow | 18 |
| 5 | 11575 | Pump, 24 VDC | 1 |
| 6 | 10135 | Tee, Union, 3/8"T x 1/4"T, PP | 15 |
| 7 | 10126 | 3/8" Stem x 3/8" Tube, Elbow | 5 |
| 8 | 10138 | Bulkhead 3/8"T X 3/8" Parker | 3 |
| 9 | 11568 | Permeate Tube, 0375, 30", BLU | 1 |
| 10 | 11565 | Inlet Tube, 0375, 27" L, BLK | 1 |
| 11 | 10886 | TQ Low Pressure Switch | 1 |
| 12 | 11567 | Drain Tube, 3/8"x30" L, GRN | 1 |
| 13 | 11564 | U-Tube,Permeate, 0375 x 9" L, Blue | 1 |
| 14 | 10136 | Elbow, .25 "T x .375" T, Parker | 3 |
| 15 | 11562 | 9" Tube, .375", BLK | 1 |
| 16 | 11566 | 6" Tube, 3/8", BLK, Inlet 2 | 1 |
| 17 | 11562 | 9" Tube, .375", BLK | 1 |
| 18 | 11690 | Cover, TQ176 - Painted | 1 |
| 19 | 11526 | Caster Wheel Kit | 4 |
| 20 | 11571 | Washer, Flat, M6 | 4 |
| 21 | 11522 | M6x14mm L, Socket Cap Head Screw | 4 |
| 22* | 71495 | Washer,Flat,3/8", Zinc Plated | 1 |
| 23 | 11570 | Connector, 0125FNPT x 025T, PP4FC2 | 1 |
| 24 | 10127 | Elbow Male, 1/4"T x 1/4" NPT, PP4ME4 | 1 |
| 25 | 11579 | Foam, TQ176 | 1 |
| 26 | 11573 | Washer, Flat M5 | 4 |
| 27 | 11572 | M5x35mm L, Socket Cap Head Screw | 4 |
| 28 | 10142 | 3.5" Tube, 3/8", BLK | 4 |
| 29 | 10144 | 3.5" Tube, 3/8", Blue | 4 |
| 30 | 10143 | 3.5" Tube, 3/8", GRN | 4 |
| 31 | 11569 | 16" Tube, 1/4", Black - Pressure Switch | 1 |
| 32 | 11525 | Logo Plate, SS, Kinetico, 5" | 1 |
| 33 | 11582 | 14-16 Terminal, Female, Insulated | 2 |
| 34* | 10146 | Prd Dcl - Labels / Logo TQ | 1 |
| 35 | 10150 | Plug, 3/8" PP | 3 |
| 36* | 11580 | DC Power Jack Cable Assy | 1 |
| 37 | 10500 | Module Asy-Plus TF 75 | 6 |
| 38* | 9386A | Prd Dcl-Labels / Serial Number Blank | 1 |
| 39 | 11576 | 24VDC Power Supply | 1 |
| 40 | 11577 | Power Cord - US | 1 |

See drawings following pages


6
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28
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3
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33
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18
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22
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12
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16
1
3/8"
3/8" INLET (BLACK)
5
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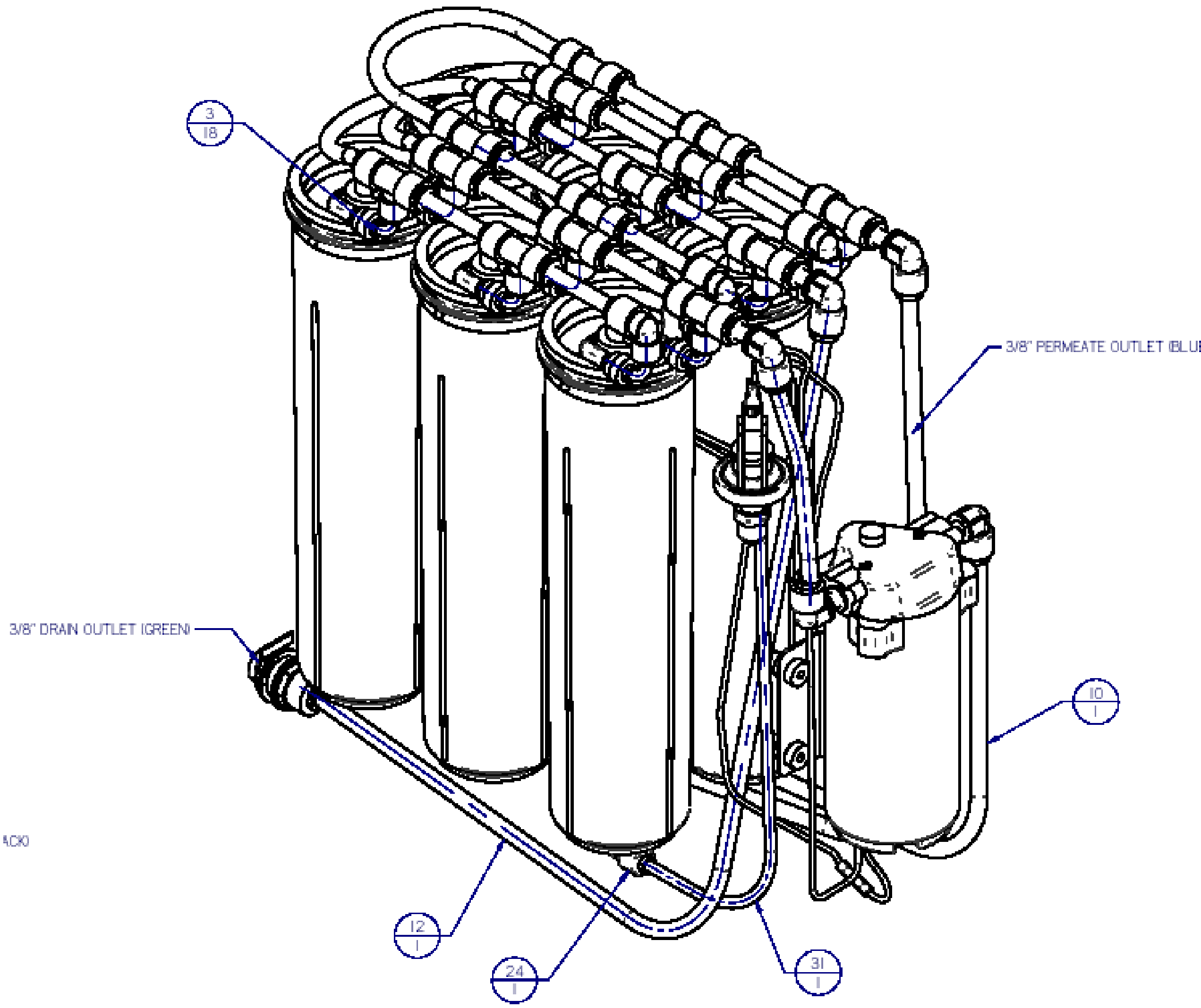
10
1
9
1


3
18
3/8" PERMEATE OUTLET (BLUE
3/8" DRAIN OUTLET (GREEN)
10
1
12
1
24
1
31
1

## PARTS

| Part # | Description | TQ Units | | | | | |
|---|---|---|---|---|---|---|---|
| | | 172 | 172n | 173 | 173n | 174 | 176 |
| 10134 | Bracket, TQ 172/173 | X | X | X | X | n/a | n/a |
| 11574 | Bracket, TQ 174/176 | n/a | n/a | n/a | n/a | X | X |
| 10148 | Cover, TQ 172/173 | X | X | X | X | n/a | n/a |
| 11561 | Cabinet, TQ 174/176 Black | n/a | n/a | n/a | n/a | X | X |
| 11578 | Cover, TQ 174/176 | n/a | n/a | n/a | n/a | X | X |
| 10500 | Module Asy-Plus TF75 | X | n/a | X | n/a | X | X |
| 10501 | Membrane, TF75 GPD | X | n/a | X | n/a | X | X |
| 10159 | Module Asy-Nano | n/a | X | n/a | X | n/a | n/a |
| 10121 | Membrane, Nano | n/a | X | n/a | X | n/a | n/a |
| 10141 | Prefilter Head, 1/4fnptx1/4fnpt | X | X | X | X | n/a | n/a |
| 10126 | Elbow 3/8T x 3/8T | X | X | X | X | X | X |
| 10135 | Tee, Union 3/8 TR x 1/4 TB | X | X | X | X | X | X |
| 10125 | Conn 1/4MNPT x 3/8 T | X | X | X | X | X | X |
| 10138 | Bulkhead Union 3/8T | X | X | X | X | X | X |
| 10139 | Elbow Tube Custom | X | X | X | X | X | X |
| 10142 | Tubing, 3/8 x 3 1/2 lg Blk | X | X | X | X | X | X |
| 10143 | Tubing, 3/8 x 3 1/2 lg Grn | X | X | X | X | X | X |
| 10144 | Tubing, 3/8 x 3 1/2 lg Blu | X | X | X | X | X | X |
| 10152B | Manual/Install TQ | X | X | X | X | X | X |
| 10886 | Switch Low PSI TQ 174/176 | n/a | n/a | n/a | n/a | X | X |
| 11575 | Pump, 24 VDC | n/a | n/a | n/a | n/a | X | X |
| 11577 | Cord, Power US 120/60 | n/a | n/a | n/a | n/a | X | X |
| 11587 | Cord, Power- European | n/a | n/a | n/a | n/a | X | X |
| 11642 | Cord, Power- UK | n/a | n/a | n/a | n/a | X | X |
| 11643 | Cord, Power- Australia | n/a | n/a | n/a | n/a | X | X |
| 11644 | Cord, Power- Denmark | n/a | n/a | n/a | n/a | X | X |
| 11521 | Wheel, Caster | n/a | n/a | n/a | n/a | X | X |
| 11708 | Kit, Cabinet Feet, Rubber | | | | | X | X |
| 11580 | Cable Assembly 24VDC | n/a | n/a | n/a | n/a | X | X |
| 11525 | Logo Plate Staninless Steel | n/a | n/a | n/a | n/a | X | X |

# SYSTEM LOG

| Date | Pressure | Permeate Flow | Temperature | QUALITY Feed | QUALITY Permeate | Comments |
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# TQ REVERSE OSMOSIS SERIES

**Manual part number: 10152B**